

Tushar Kumar Mahata

Machine Learning Engineer — Deep Learning · Computer Vision · NLP

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I turn deep learning research into things people can actually use — **image recognition, object detection and NLP**, taken from raw dataset to deployed endpoint. Six years of freelance engineering (2020–present) taught me to ship under real constraints: real clients, real deadlines. **Google-certified TensorFlow Developer**. I care most about the unglamorous half of the job — preprocessing, tuning and the API in front of the model, the part that decides whether it survives contact with production.

EXPERIENCE

WordPress & Shopify Developer

2020 — Present

Fiverr · Freelance

- Maintain, fix and **customise** WordPress and Shopify builds for a continuous roster of clients.
- Take designs from concept to live storefront — full **theme design and redesign**, not just tweaks.
- Hunt down slow assets and bloat, then apply **speed optimisation** so pages actually load.
- Run **on-page SEO** passes — structure, metadata, content — so client sites get found.

SELECTED PROJECTS

Dog's Breed Classifier

95%
ACCURACY

FINE-GRAINED IMAGE CLASSIFICATION

Trained a transfer-learning model to tell 120 dog breeds apart using the Stanford Dogs dataset from Kaggle. Built a TensorFlow input pipeline for efficient image classification, then pushed architecture and hyperparameter choices through a long experiment loop to land at 95%. Shipped as a **Dockerised FastAPI service** with endpoints for image upload and live prediction.

TensorFlow

Transfer Learning

OpenCV

FastAPI

Docker

Footballer Classifier

81.32%
VAL. ACCURACY

673
IMAGES

OBJECT DETECTION & ML PIPELINE

Find and name famous footballers in photos. Faces located with Haar Cascade, cropped, then pushed through **wavelet transformation** for feature engineering before a spread of ML models ran through a single pipeline — best performer hit 81.32% on validation. Deployed on Flask so anyone can drop in an image and get a prediction back.

OpenCV

Flask

ML Pipeline

Scikit-Learn

HTML / CSS

NLP · RESEARCH REPRODUCTION

Rebuilt the model architecture from the original paper for sequential sentence classification over the PubMed 200k RCT corpus: rigorous preprocessing, several competing architectures, disciplined hyperparameter tuning, and **interpretability work so the results can be explained** — not just reported — for biomedical literature classification.

TensorFlow

NLP

Transfer Learning

Matplotlib

Explanatory Analysis

SKILLS

LANGUAGES

Python

C

SQL

HTML

CSS

ML & FRAMEWORKS

TensorFlow

Keras

Scikit-Learn

FastAPI

Flask

CORE DISCIPLINES

Image Recognition

Object Detection

Transfer Learning

Predictive Maintenance

Model Optimisation

APPLIED ML

GenAI Models

NLP

Real-time Data Processing

GPU Programming

DATA & TOOLING

OpenCV

Matplotlib

Docker

Relational Databases

Statistics

ALSO

WordPress

Shopify

On-page SEO

Speed Optimisation

Team Collaboration

EDUCATION & CERTIFICATION

BACHELOR'S DEGREE

Computer Science & Engineering

Daffodil International University, Dhaka, Bangladesh
2016 — 2021 · CGPA 3.10 / 4.00

CERTIFICATION

TensorFlow Developer Certificate

Google · Issued September 2023
Valid through September 2026